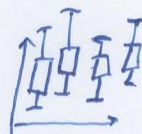


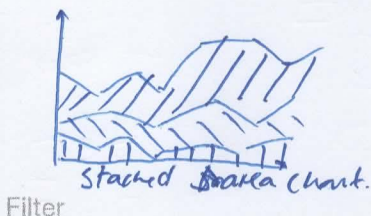
Symbol map for
train/tram stations.



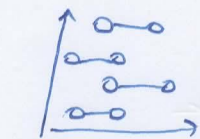
Heat map for
times/locations.



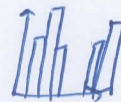
Box & whisker, show
usage behaviours.



Try compare 2 diff.
modes.



Dot plot, compare
changes across COVID period



Histogram for
mode usage patterns.



Tree map for each
mode.

→ Treemap might be hard to read as certain transport modes will dominate others e.g. regional bus vs. metro bus.

→ Also limited no. of modes which are present (not visually appealing)

→ Box and whisker plots are very useful for showing statistical info (e.g. quartiles, mean etc.) but may be too difficult to explain. Better suited to academic environments / more in-depth analysis.

Categorise

→ All public transport modes in Victoria.

→ Compare against each other.

→ Passenger numbers.

→ Punctuality.

→ Number of stops.

→ Busiest times.

→ Deep-dive on certain modes

→ e.g. Choose train, bus, tram

→ Fine details.

→ Which stations are busiest?

→ Why?

→ Trends for school holidays.

→ Train dataset has very high fidelity

Combine & Refine

→ Pie chart could possibly be combined with close-up map to show splits for stations of interest. Would ~~not~~ require lots of space + limited entries on map.

→ Dot plot could be modified to have the average be the mid-point and the dots show deviation from the mid-point (mean)

→ Shows similar info. to box + whisker plots but with less confusing layout and less information.

→ Combine stacked chart + histogram to show frequency of multiple modes over time

Summarise and question

→ How can I incorporate interactivity?

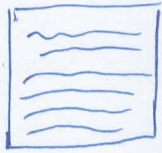
→ Use legend filtering, sliders, dropdown menus, brushing for area chart.

→ Will map be too cluttered?

→ Maybe, look at using zoomed context map to further examine areas on map.

Public Transport in Victoria

Overview of Public transport (usage)



Trams



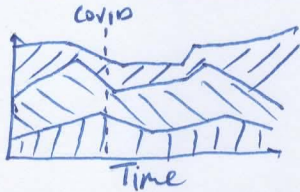
Trains



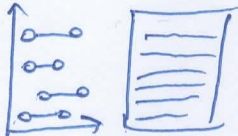
Buses.

Annual / monthly passenger numbers

Trends during COVID



"Deep-dive" on trains



School holidays effect

Part / Focus



→ Focus on the 'big picture' whole transport system first.

→ Comparisons between each of the different modes (separate for clarity).



Mode 1



Mode 2



Mode 3.

→ Shift focus to one particular mode, should have lots of information.

→ Good candidate is the train system

→ Use map to show which stations are the busiest.

→ Interact with the train lines on each of the charts, e.g. connected dot plot.

Sectioned layout

Components / Operations

→ Year slider:

2016

2024

Way less clutter than showing each year on its own chart, easy to implement.

→ Brushing on stacked area chart:



Allows user to zoom in on period of interest. Can be implemented in Vega.

→ Tooltip interactions on map:



Adds additional context, clarifies any ambiguity.

Pro & Cons

→ Separating modes into individual charts can add a lot of clutter to the page, might be better using interactivity

→ Clear hierarchy of information due to headings.

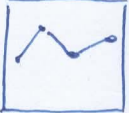
→ "Drill down" approach from top-down is intuitive + easy to follow.

→ Need to ensure the height of the page isn't too large (will negatively impact readability)

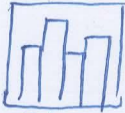
Public Transport in Victoria

Select mode: Bus

Station location



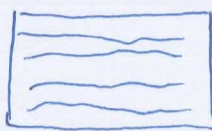
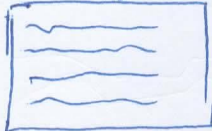
Punctuality



Passenger



AM vs. PM

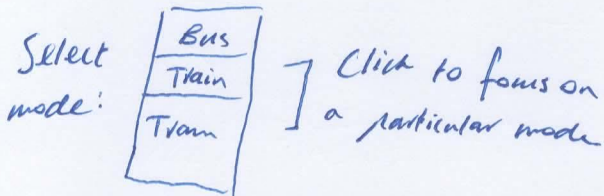


Split paragraphs

Part / Focus

→ Dashboard layout which seeks to show a large amount of info. about selected transport mode.

→ Graphs/charts update after user selection.



→ Paragraphs update to provide dynamic analysis about selected mode.

→ Would probably require JS to implement.

Dashboard layout

Components / Operations

→ Main operation is selecting mode from a drop-down.

→ This updates the dashboard to display info. about the selected mode.

→ Other operations would also be included, such as:

→ Legend filtering

→ Year selection slider

→ Brushing with click + drag operation.

Pro & Cons

→ Information is easily readable, density presented.

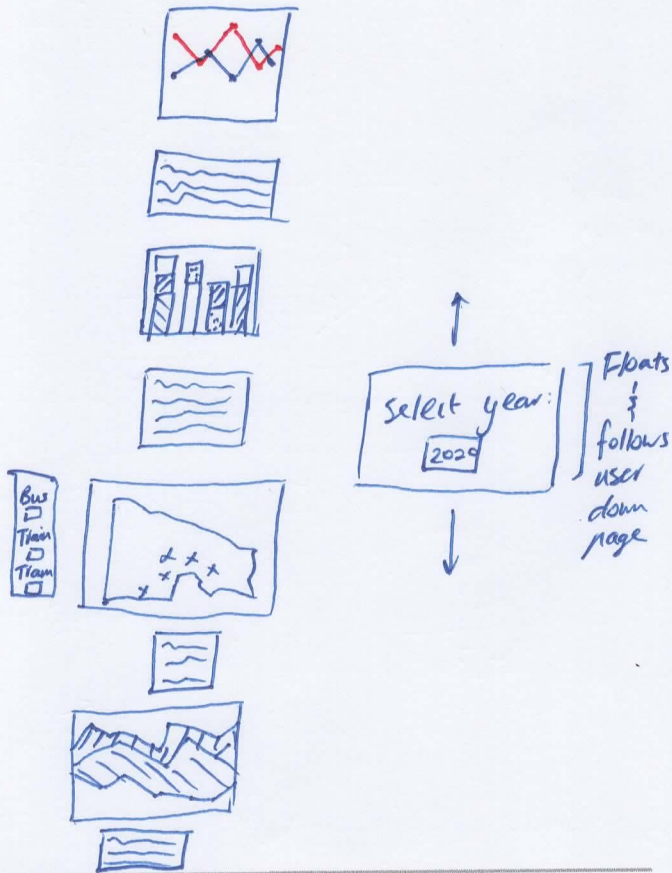
→ Updating charts/maps/graphs is easy with Vega.

→ Updating annotations & paragraphs would be significantly more difficult.

→ Comparisons between the different modes would be unclear / difficult to incorporate

→ Could add a separate section for comparisons (that's static) below the dashboard.

Public Transport in Victoria



Part / Focus

→ Main focus of this layout is to see the evolution of various metrics relating to transport (passenger, punctuality, weekday) over time.



→ This is accomplished with the floating year selector.

→ Unlike individual selectors for each chart, this unified approach streamlines the year selection process.

Name Lyndon Butman
Date

Title

Description

Article layout

Components / Operations

→ A floating year selector will follow the user down the side of the page as they scroll.

→ This acts as a global filter for all relevant charts.

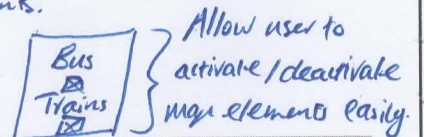
→ Other standard interactivity options will be implemented:

→ Legend filtering,

→ Brushing

→ Dropdown.

→ Selection boxes for map to add / remove elements.



Pro & Cons

→ Top-down article layout allows users to be closely guided through content.

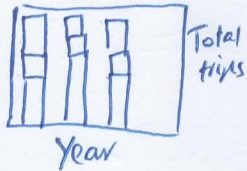
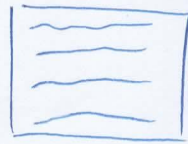
→ Hierarchy is determined by order of elements.

→ Users will have to scroll to engage meaning not all information is ~~readable~~ readable / accessible at once.

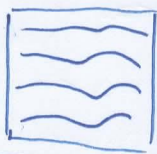
→ Article layout lends better to guiding through charts, might discourage users from exploring the content.

Public Transport in Victoria

General usage (all modes)



Punctuality



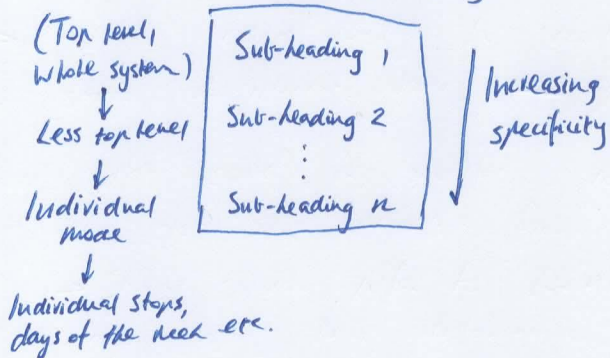
Trains (deep dive)



Parti / Focus

→ The focus of the vis changes and flows with each subsection.

→ Generally though, each progression through a heading downwards represents an increase in the specificity of the info:

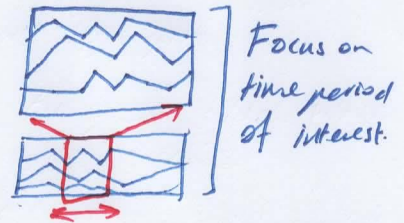


→ The use of a number of operations allows the user to meaningfully interact with the vis and explore the data in detail.

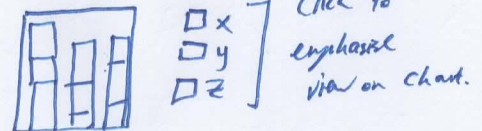
Sectioned / dashboard

Components / Operations

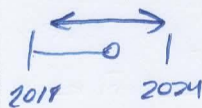
→ Brushing on stacked area chart



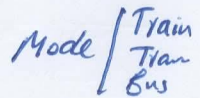
→ Legend filtering



→ Year slider



→ Mode selector



→ Tooltips

Details

→ This approach combines the section layout with mini dashboards in each section

→ Starts off with top-level information / trends e.g. total trips, split by mode

→ Transitions to top-level for train mode

→ Drills down even further to station / train line level.

→ Top-down hierarchy to guide viewer whilst still providing freedom to explore.